

Network Networking

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MCS Program

Topic Coverage

01 Network Foundations

The Internet and Network Fundamentals

02 Network Edge && Network Core

Explain the difference between

03 Communication and Network Protocols

- What a network protocol is ?
- why protocols are necessary?

04 Understanding Network Performance

The major factors that affect network performance

05 Network Security

Why security is essential in modern computer networks?



01

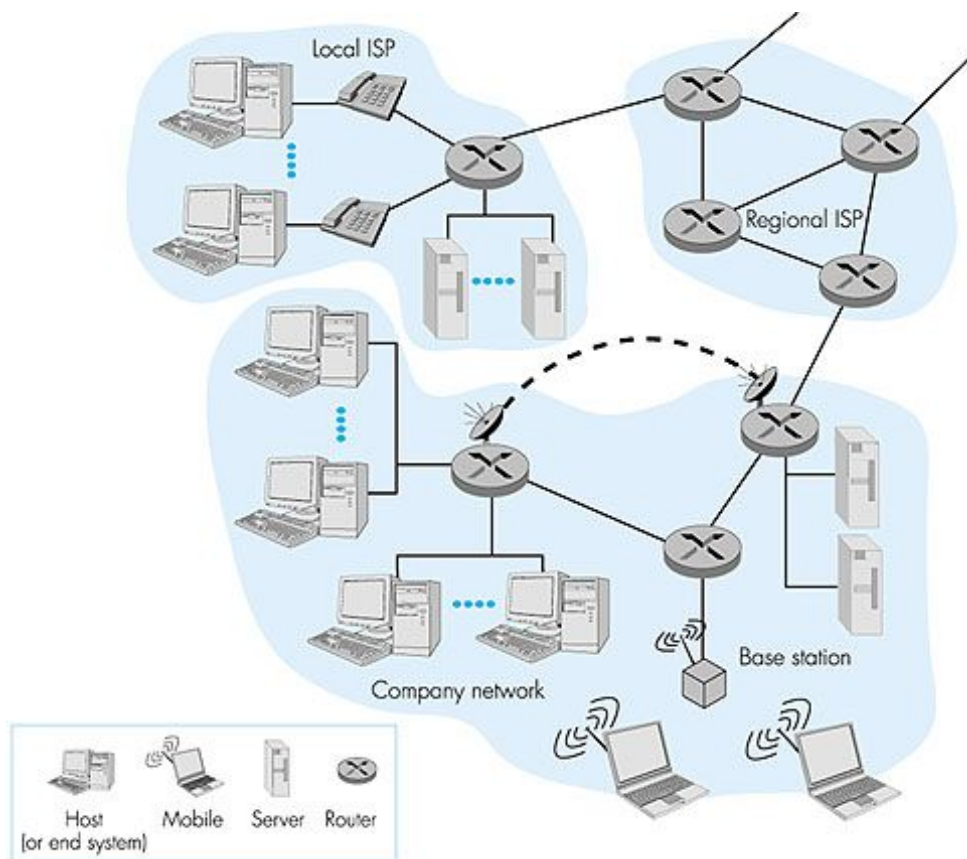
Network Foundation

“Computer Network is a group of two or more computers or devices that are connected so they can communicate and share resources, such as files, printers, applications, or internet access.”

Network of Networks

The Internet is the ultimate **network of networks**, meaning it is a massive global system made by joining smaller, independent computer networks together.

- The Internet is commonly called a “**network of networks**” because it is made up of many separate computer networks connected together.
- Billions of devices can communicate because individual networks are connected together, and routers use shared protocols and addresses to direct data packets to the correct destination.



End-System or Hosts

Sample of devices and how they can connect together

Laptops & Desktops

- Clients / Nodes
- Full TCP/IP stack (Wi-Fi, Ethernet)
- General-purpose computing, web browsing, local software execution.

Smartphones & Tablets

- Mobile Clients
- Full TCP/IP stack (Cellular, Wi-Fi)
- On-the-go media access, app-driven communication, location services.

Physical Servers

- Host / Provider
- High-throughput TCP/IP, fiber interconnects
- Dedicated compute engines hosting websites, databases, and enterprise services.

Cloud Systems (VMs/Containers)

- Virtual Hosts
 - Software-defined networking (SDN)
 - Virtualized server instances running on top of physical data center infrastructure.
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Connect through Wifi Routers

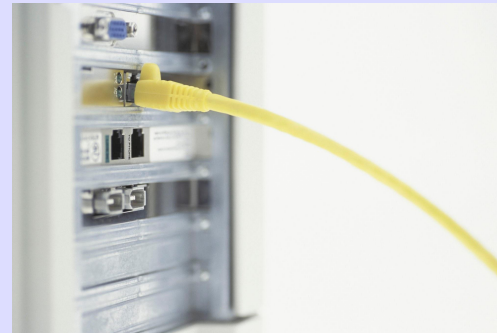
At Home, School, Library, e.t.c

Real-World
Example



Connection through
cellular Signal

*Long range normally
know as 5G - 6G signal*



Ethernet/Fiber

*Stable and
High-bandwidth*

02

Network Edge && Core

Network Edge

Hosts (End Systems)

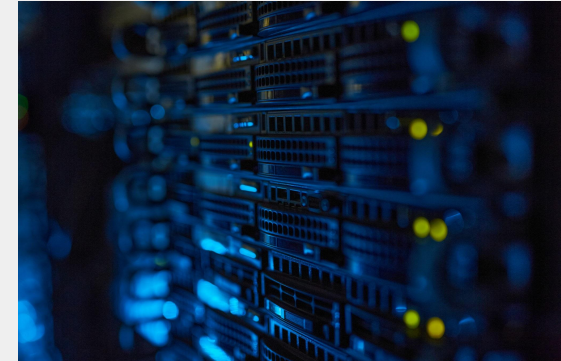
The network edge is the part of the network architecture where users and end devices interface with the Internet. It is decentralized, diverse, and focuses on "access" and "local service."

Clients

- Devices that request information or services.
- *Examples: Laptops, smartphones, PCs, wireless tablets, IoT devices (smart thermostats, security cameras).*

Servers

- Powerful machines that store data and provide services to clients. While many servers live in massive data centers deep within the network core.
- *Examples: Web servers, database servers, application servers.*



Access Network

An access network is the physical link that connects a host to the first router (also known as the "edge router") of the ISP network. The access network is often the bottle-neck in terms of bandwidth.

Home Networks

- Provides Internet access within a residence.
- Home Router (Combines router, switch, firewall, and wireless access point), Modems (DSL/Cable/Fiber), Wi-Fi access.

Enterprise Networks

- Connects users within a company or university setting.
- Ethernet Switches, Wireless Access Points (APs), Enterprise-grade Firewalls, Internal Servers.

Mobile (Wide-Area) Networks

- Uses cellular technology to provide wireless access over large areas.
 - Base Stations, 4G/5G Radio Access Network (RAN), Mobile devices.
-

Network Core

Internet (Backbone)

The network core is the "backbone" of the Internet. Its sole purpose is to provide high-speed, reliable transport of data packets over long distances, connecting many different access networks together.

Routers (Core Routers)

- Specialized, high-capacity routers that interconnect standard access networks and other ISPs.
- They focus on routing packets between different networks rather than connecting directly to end devices

Switches (Core Switches)

- High-performance switches used within data centers or between routers to manage high-speed physical connections.

Links

- The physical medium connecting these devices, primarily high-capacity fiber-optic cables



Packet traveling methods

How data transferring in network

Packet Switching

- Data is broken into small, discrete blocks called packets. Each packet contains its destination address in its header. Routers examine the header of each packet independently to determine the best outgoing link
-

Used by the Internet

- No dedicated path or reserved bandwidth. Bandwidth is used on demand (statistical multiplexing). This is more efficient for "bursty" traffic, such as web browsing or email.
-

Circuit Switching

- A dedicated, end-to-end path (a "circuit") is established between the sender and receiver before any data is transmitted
-

Used by Legacy Telephone Networks

- Bandwidth is reserved exclusively for that connection for its entire duration. While this guarantees performance, it is highly inefficient if the connection is idle.
-

Show a Diagram

03

Communication &
Network Protocol

Network Protocol

A **network protocol** is an agreed-upon set of rules, formats, and conventions that govern how data is formatted, transmitted, routed, and received across interconnected systems.

HTTP/HTTPS

- Purpose: Defines how messages are formatted and transmitted for fetching web resources.
- Participation Level: Operates at the Application Layer.

TCP

- Purpose: Provides reliable, ordered, and error-checked delivery of a stream of octets between applications running on hosts.
- Participation Level: Operates at the Transport Layer.

IP

- Purpose: Responsible for addressing nodes and routing discrete data packets
- Participation Level: Operates at the Network Layer

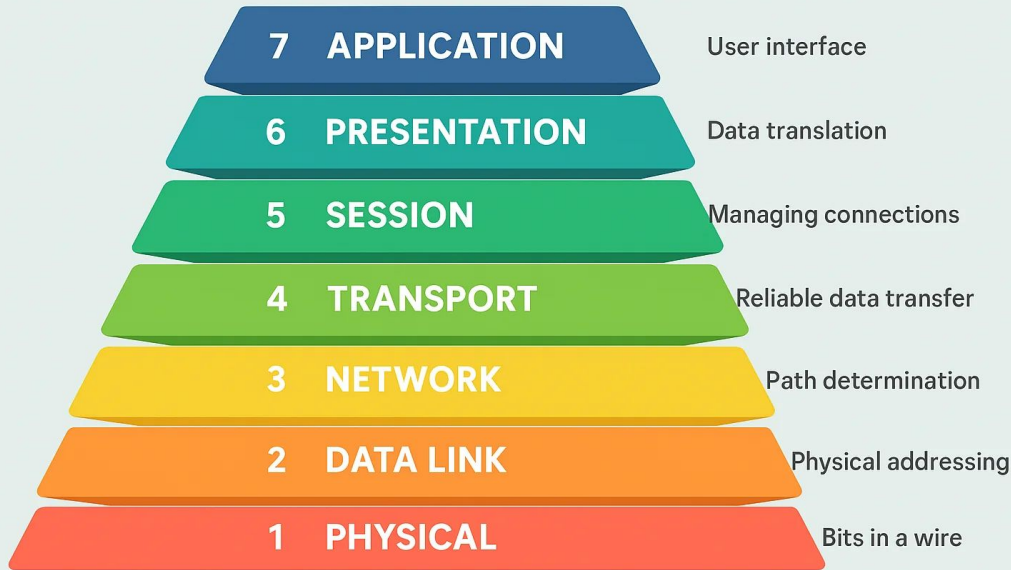
DNS

- Purpose: Translates human-readable into machine-routable IP addresses
 - Participation Level: Operates at the Application Layer.
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Network Protocol

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THE OSI MODEL



8 Popular Network Protocols

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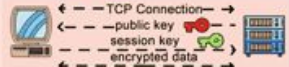
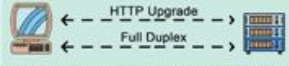
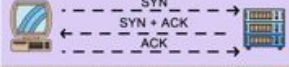
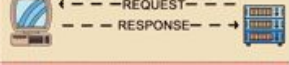
Protocol	How does It Work?	Use Cases
HTTP		 Web Browsing
HTTP/3 (QUIC)		 IoT Virtual Reality
HTTPS		 Web Browsing
WebSocket		 Live Chat Real-Time Data Transmission
TCP		 Web Browsing Email Protocols
UDP		 Video Conferencing
SMTP		 Sending/Receiving Emails
FTP		 Upload/Download Files

Diagram + Sample real interact

Explain how several protocols work together to make that activity possible.

04

Understand Network Performance

Affect network performance

01 Throughput

The actual rate of data successfully delivered over a network path per unit of time

02 Bandwidth

The theoretical maximum data-carrying capacity of a network link, measured in bits per second

03 Delay

The total time required for a data packet to travel from the source node to its destination

04 Transmission Delay

The time required to push all bits of a packet onto the physical transmission medium

05 Queuing Delay

The time a packet spends waiting in a router or switch queue buffer prior to being processed and transmitted

06 Packet Loss

The percentage or ratio of transmitted data packets that fail to reach their destination

Diagram + Sample real interact

One simple network observation or demonstration

05

Network Security

Security Concepts

01 Authentication

The process of verifying the identity of a user, device, or process before granting network access

02 Encryption

The mathematical process of encoding plain text into cipher text using cryptographic keys

03 Confidentiality

A fundamental pillar of the Security Triad (CIA) that ensures sensitive data is accessible only to authorized entities

04 Network Threats

Any potential vulnerability or deliberate malicious action aimed at disrupting operations, stealing data, or gaining unauthorized access to network resources

05 IP Spoofing

A technique where an attacker alters the source address in the IP packet header to masquerade as a trusted network entity

06 Packet Sniffing

The act of capturing and analyzing raw data packets traveling across a network using packet analysis tools

Network Defend Technique

Network security protects data integrity, preserves system availability, and shields sensitive information from unauthorized access.

Firewalls

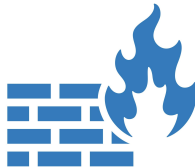
- Perimeter Control & Filtering
- Inspects incoming and outgoing network traffic based on predefined security rules to block unauthorized connections and malicious protocols.

Virtual Private Networks (VPNs)

- Encrypted Tunneling
- Establishes an encrypted point-to-point connection over public networks, masking IP addresses and protecting data from packet sniffing.

HTTPS (TLS Encryption)

- Transport Layer Protection
 - Encrypts HTTP communication between web browsers and servers using Transport Layer Security (TLS), mitigating eavesdropping and man-in-the-middle attacks.
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Real World Example

Scenario: Consider a corporate employee connecting to an unencrypted public Wi-Fi network at an airport:

The Attack

- An attacker on the same Wi-Fi network launches a packet sniffing tool to capture local airwave traffic. The attacker simultaneously uses IP spoofing and ARP cache poisoning to position themselves between the employee's laptop and the router, executing a Man-in-the-Middle (MitM) attack.

The Risk

- If the employee accesses an internal company application via unencrypted HTTP, the attacker intercepts session cookies and login credentials in clear text.

The Mitigation

- By enforcing HTTPS and routing all laptop traffic through a VPN, the transmission is encrypted end-to-end. Even though the attacker captures the physical packets via sniffing, the payload appears as unreadable ciphertext, preserving confidentiality and maintaining network integrity.
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Note: This project have been supported by AI in summarize. Check information accurate and rephrase

THANK YOU